श्री माँ

(1)

Artificial intelligence

Introduction :-

" Can machines Think" ?

1. Smart phone. 'Siri' with in the Second Find at ASWAR-
2. Alexa / Find out Answar
3. Smart ~~Cas~~ Cars 50,000 car in ~~USA~~ USA tasla
   ↓ with out driver
4. machins think.
5. ~~Dasian~~ Decision power.
6. ~~Source~~ Source any vidio
7. Pizza order.
8. colastral value high.
9. Phone Link.
10. AI data - Face ~~Ideti~~ identifi
    = Reasoning : 40% marks
    = Learing. - ~~expers~~. exprieins.
    - problem Solving. Game problem.
    - perception. / tasala company

(2)

AI

1. mind use. machine
2. Set programm acoding to human.
3. Computer Programm / Reminder.
4. machine Learing.
5. 1955 ~~JOH~~ John MC
6. Robot.
7. Smart phone. Siri.
8. Google map.
9. worker use. Factrics.
10. vidio Games.

CS

(2)

Uo

Benefities.

1. error remove.
2. Fast work. / fast desion.
3. No Need refreshment.

dis Benifities.

many.
Job.
robot mind

Applications of AI

Scope of the AI

1. Natural Language processing (NLP)

The long standing goals of artificial intelligence is the creation of programs that are capable of understanding and generat ~~fun~~ human language.

2. Expert systems :- They provides expert specialized areas. Expert system have built military system like aircrafts.

3. Game playing

Games generate extremely large search spaces. These are large and require power ful techniques for determining are alternative to problem space. These techniques are called heuristics

4. Computer vision :-

The world is composed of three dimension (3-D) objects, but the inputs to the human eyes and computer TV cameras are two dimensional (2-D). At present, there are only limited ways of representation 3-D information directly they are not as good as what human evidently use.

5. Theorm proving :-

Theorem proving ~~rea~~ research was responsible for much of the early work in formalizing search algorithms and developing formal representation Language

6. problem Solving :-

PROVERB (1999) is a computer program that solves crossword puzzles. A Large database of past puzzles and a varity of information sources Including dictionaries and online database.

7. medical diagnosis :-

AI can be used in various manners. medical system might scan laboratory test results. Send reminders or warning through an email system. When patient's case is complex, rate of the person making the diagnosis is simply inexperienced ~~an expert system~~ diagnoses based on patient's data. many medical images can now be automatically from plan X-rays, CT and MRI scans.

8. Robotics and planning :-

Robots design could perform tasks with some ~~der~~ degree of flexibility and responsiveness to the outside world. robot that is capable of performing certain ~~atomic~~ atomic actions. It is higher Level task such as moving across an obstacle ~~o~~ filled room.

~~Medical Diagnosis in~~

a machine Learning :- expert system may perform extensive and costly computation to solve a problem. If expert system is given the same or similar problem a second time. is usually does not remember the Solution.

AI provide the platform to make a machine Can Learn. It increases the performance of our expert system.

10. AI in E-Commerce :-

AI approaches are useful in the development of E-Commerce system. B2B and B2C two system. AI Focused on B2C transaction. AI technologies used mainly for supple chain / menagement.

11. AI In E-Tourism :-

In the tourism Field, it is possible to find travel agencies, transportation, hotel or car companies that offer different products on the web.

It is possible to find bargain prices only for users that buy the web. tourism Field has been modified the electronic transactions.

Information stored in the web.

Some of those problems are.

(a) We have to search a lot of websides to get useful information.

(b) The information served by the comanies could change along time.

(c) The information sources is not always possible etc.

It is necessary to find new approaches to allow design and build system automatically

- Search and retrieve useful information.
- manage and reason with different solution
- Gain efficiency using the Learned knowledge by other agents.
- Adapt to profile user to customize the answer.

- AI Fiance -
- AI in Data ~~Secrit~~ Secure -

Approaches to AI

1. Hard or Strong AI :-
Strong AI aims to build machines that can truly and solve problem. Strong AI maintains that suitably programmed machines are capable of connitive mental states.

2. Soft or Weak AI :-
Weak AI deals with the creation of some form of computer-based artificial intelligence soft ware that cannot truly reason and solve problems.

3. Applied AI :-
Applied AI aims to produce commericially feasible "smart" system such as security system face recognize who are permitt to enter a particular building.

(4) cognitive AI :- (Theory of mind) (10)

Computers are used to test theories about how the human ~~mai~~ mind work ~~throg~~ the use of computational models.

The importance is of how it is done but not of what is done.

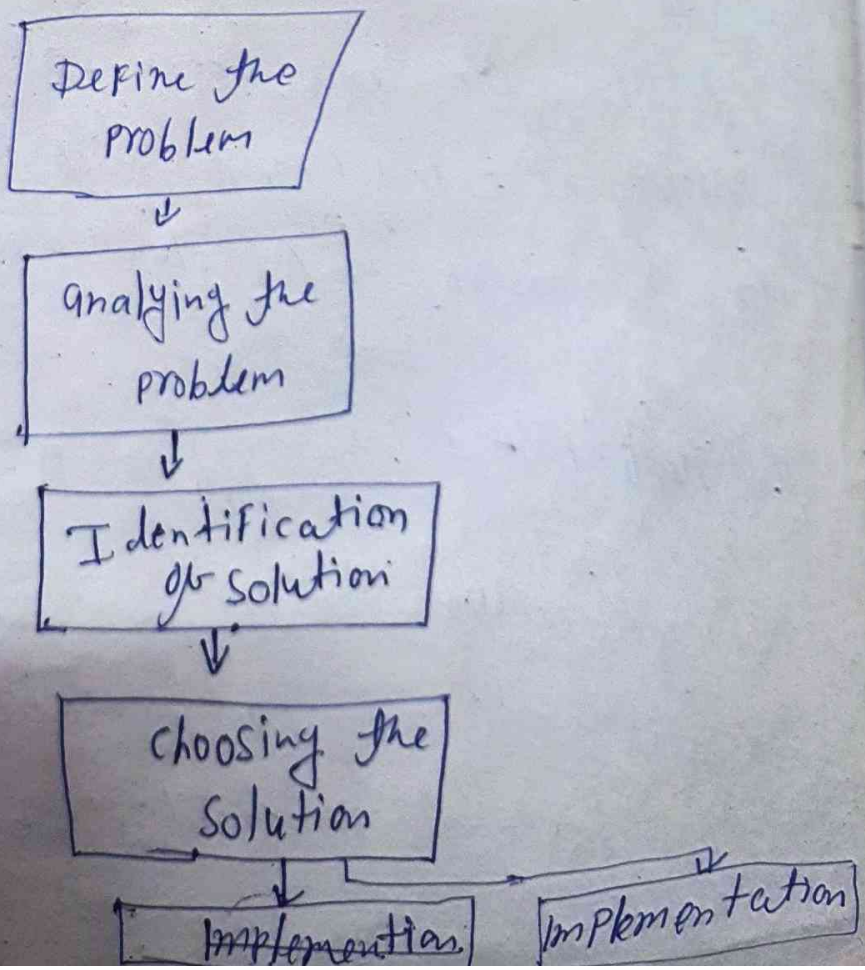
The program must operate in an Intelligent manner.

(5) Reactive machine - (7) Theory of mind.

(6) Limited memory (8) ~~Self~~ ~~Self~~ Self-Awareness

problem Solving :- - Sudoku

AI - Agent



problem Searching –

– In genral, Searching refers to as finding information one needs.

– Searching is the most commonly technique of problem solving in A.I.

~~– Dia~~

– Define the problem.

– analying the problem

– Identification of the solution.

– choosing the solution.

– Implementation –

State Space Search –

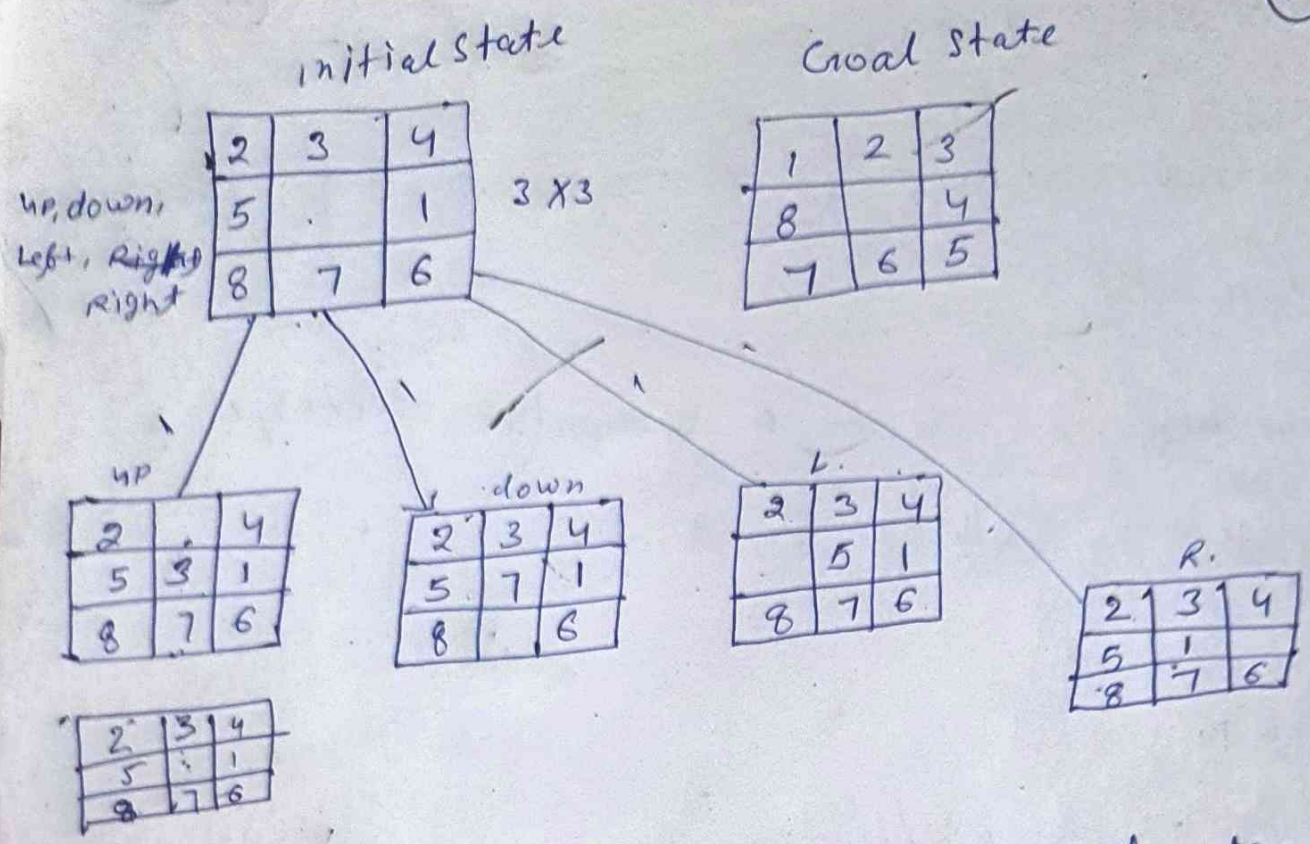
– precise – How to Represent the problem.

– Analyze – 'state space search'

S {S, A, Action (S), Result (S, a), cost (S, a)}

S = Start, Goal, ~~intermidetedi~~ intermidiat state

action, Set of action



states :- A state description specifies the location of each of the eight tiles and the blank in one of the nine squares.

~~Successor~~

Intitial state :- any state can be designated as the Intitial state.

Successor function :- This generates the legal states that result from trying the four action (blank moves Left, Right, up or ~~dow~~ down

Goal Test :- This checks whether the state matches the goal configuration as shown in figure above.

Path Cost: Each step cost is 1, So the path cost is the number of steps in the path.

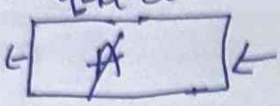
Uninformed Search

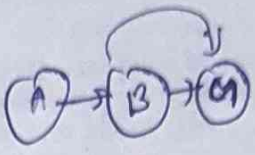
Uninformed Searching	Informed Searching
1. Search without Information	Search with Information
2. No knowledge	use knowledge to find steps to solution.
3. Time Consuming	Quick Solution.
4. more complexity (Time, Space)	Less complexity (Time, Space)
5. DFS, BFS etc.	A*, Heristic (হিউরিস্টিক) DFS, Best First Search

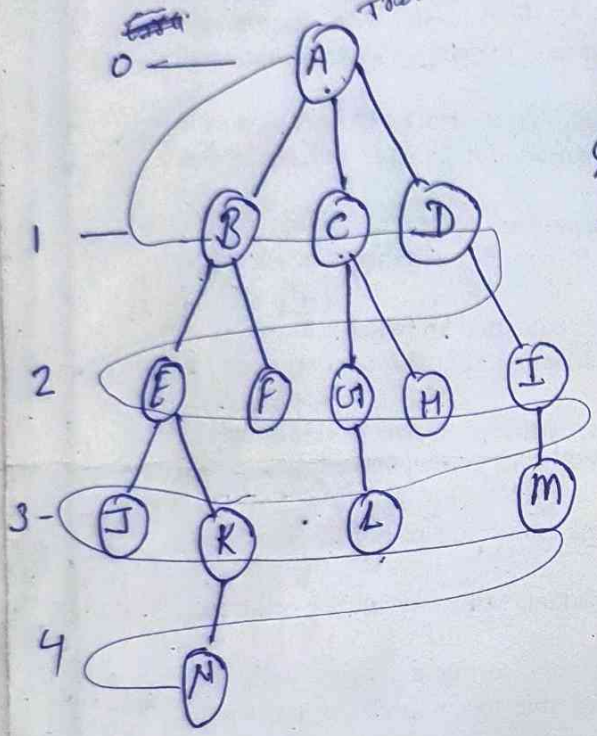
Breadth first Search (BFS)

1. uninformed Searching .. Blind technique.
2. FIFO (queue)
3. Complete.
4. optimal - short cost
5. Time Complexity

queue database







~~B~~ C D
~~C~~ ~~D~~ E F
~~D~~ E F G H
~~E~~ F G H I
~~F~~ G H I J K
~~G~~ H I J K
~~H~~ I J K L
~~I~~ J K L
~~J~~ K L M
~~K~~ L M
~~L~~ M N
~~M~~ N
N

6. Label Search tenic.

$O(b^d)$

$\boxed{A\ C\ G}$

b - Branch Factor

d = depth -

1. ~~MILICIOUS APPLICATION~~

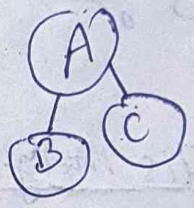
2. ~~TCP/IP PROTOCOL~~

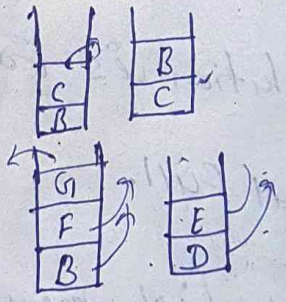
3. ~~TUNNELING~~

15 ~~30~~

DFS Depth - First Search

- Uninformed Search techique.
- stack (LIFO)
- Deepest Node.
- Incomplete
- Non optimal.
- Time complexity.



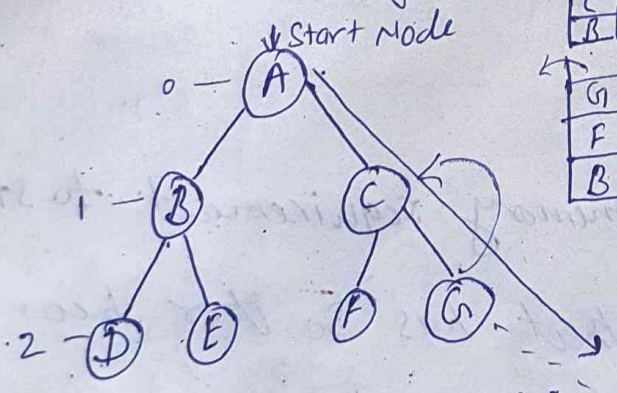


$O(b^d)$

b = branching factor

d = depth

$2^2 = 4$



A C G F B E D

A C G

Advantage of BFS

1. BFS will not get trapped exploring blind alley.
2. If there is a Solution, BFS is guaranteed to find it.
3. If there are multiple solutions, then a minimal Solution will be found.

disadvantages :-

1. many nodes may need to be explored before a solution is found, especially if the tree is ~~full~~ full.
2. It ~~need~~ need high memory requirement to store the nodes of tree that has so far been generated.

Iterative deepening Searching.

· ~~Iterativ~~

— Combination of BFS + DFS

Adv. BFS

1 it will find Goal Node any way.

dis BFS

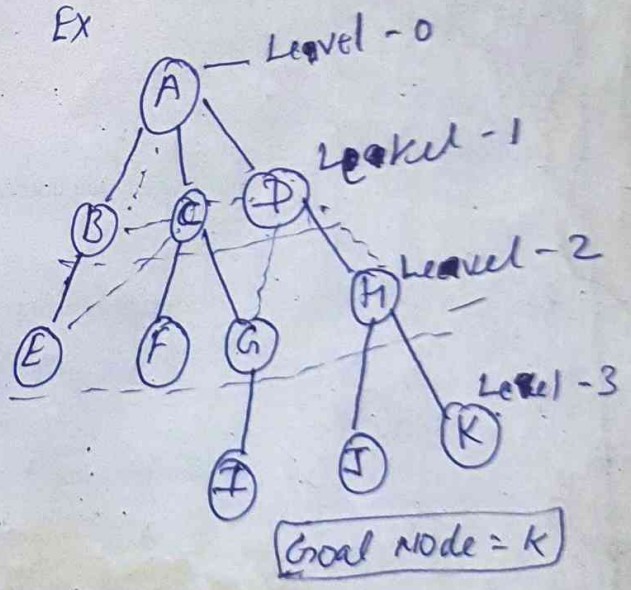
= It consumes very much ~~memory~~ memory

- Adv. DFS.

— It consume very Less memory.

Depth Level -	Interative delping Search.
0	A
1.	ABCD
2.	ABECFGDH
3.	ABECFGIDHJK

Ex

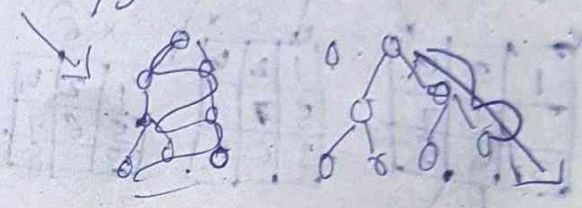


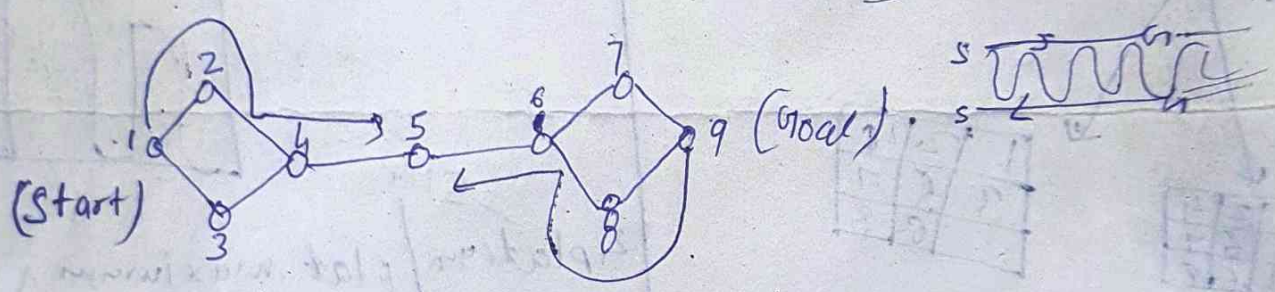
Bidirectional Search

- Two simultaneous search from an initial node to goal from goal to initial, stopping when two meet.

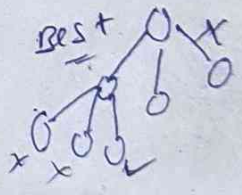
o = order of

- most Time complexity: $2(b^{d/2})$
- Complete in breadth first search / If we use BFS - Goal search - If we use DFS may be
- Not in depth first search





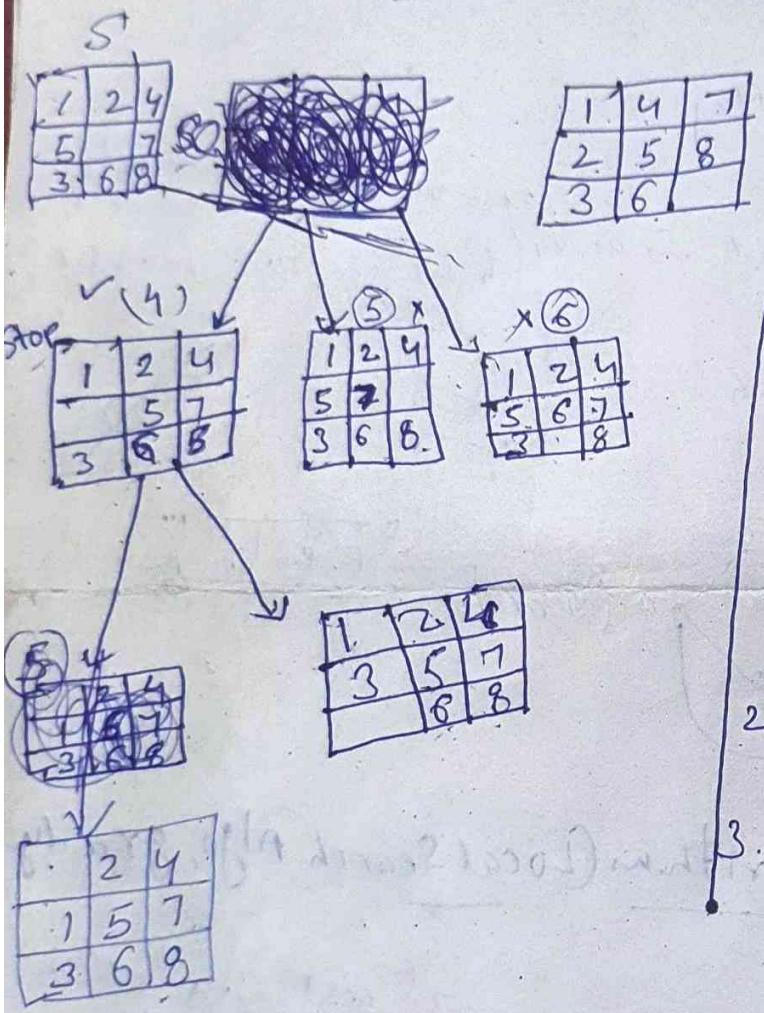
Simple Hill climbing Algorithm (Local Search Algo, greedy approch, No backtrac



1. Evaluate the initial state.
2. Loop until a solution is found or there are no operators left.
   - Select and Apply new operator.
   - Evaluate the new state.

If goal then quit.

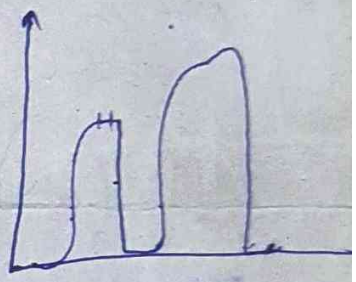
If better than current state ~~then~~ then it is new current state.



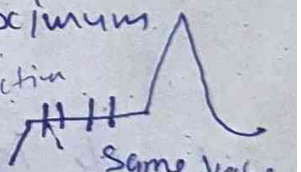
(4)

problems in Hill climbing

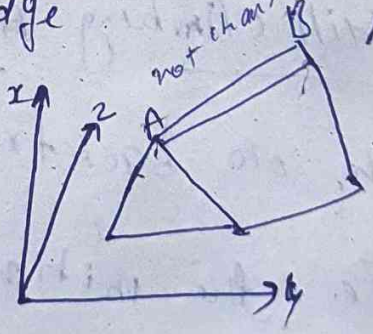
1. Local maximum

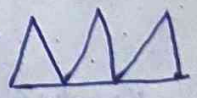


2. plateau / flat maximum → stop



3. Ridge



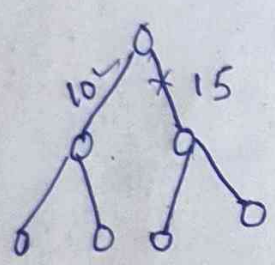


Informed Search techniques :-

Heuristic

Heuristic in AI (Rule of Thumb) [What, why, How]

- it is a technique designed to solve a problem quickly.

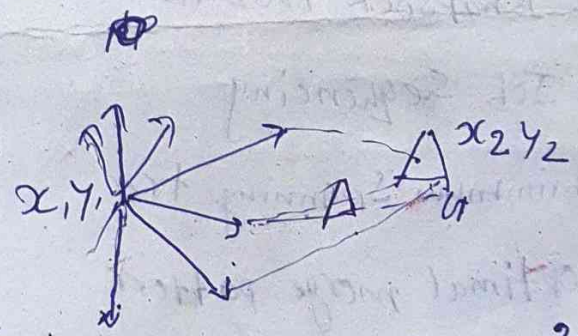


8 Puzzle $0 = (b)^d$

15 Puzzle 10^{13}

- Reduce Time

Euclidia

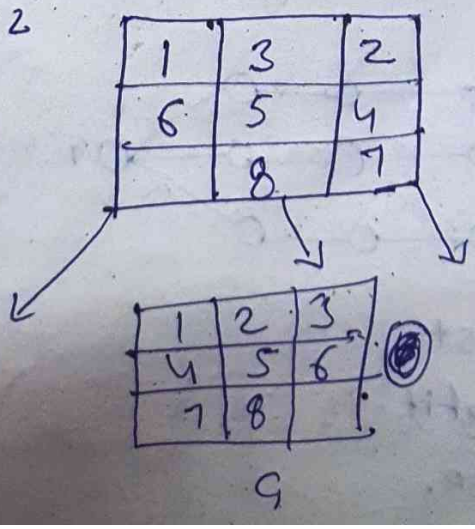


- good
- optimal for sour.

distance $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

manhaten

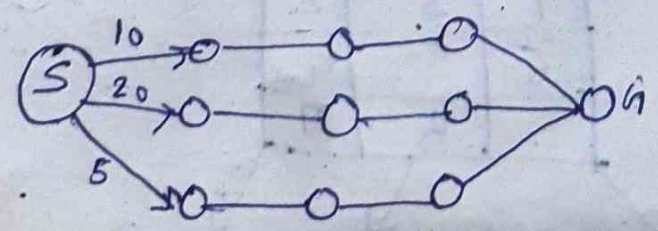
0+1+1+2+0+2+2+0+



Greedy Technique (Algorithm)

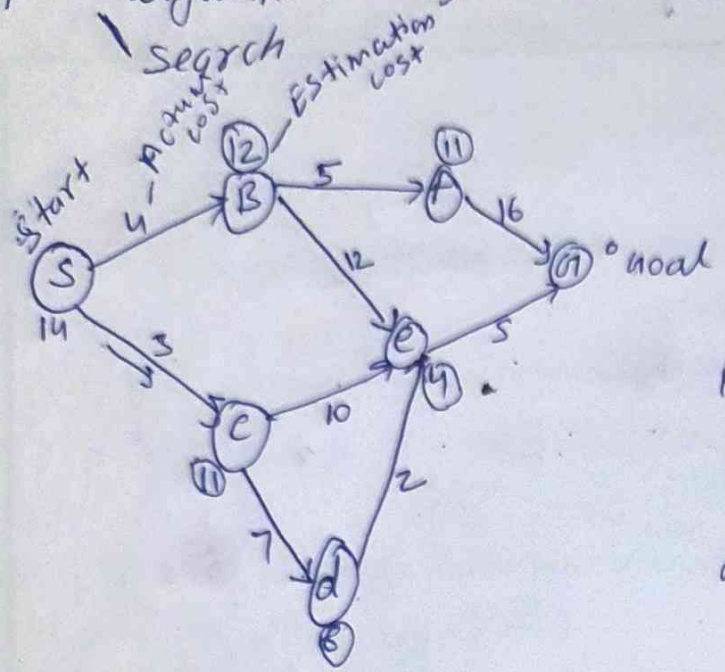
- Follows local optimal choice at each with intend of ~~finding~~ global optimum.

care
- Feasible Solution (selection)
- Optimal Solution
- Application:
  1. Knapsock problem
  2. Job Sequencing
  3. minimum Spanning tree
  4. optimal merge pattern
  5. Huffman Coding
  6. Dijkstra's algo



→ min. cost
- max profit.
- min Risk.

A* algorithm - Informed Searching
search



cost found

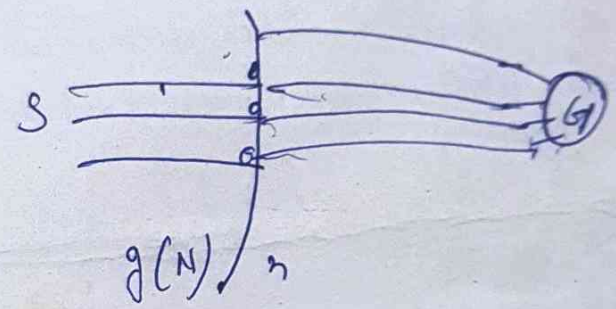
$F(N) = g(N) + h(N)$

Actual cost from start node to n

Estimation cost from n to Goal node

$O = (b^d)$

Branch Factor



$F(S) = 0 + 14 = 14$

$S \to B$

$4 + 12 = 16$

$S \to C$

$3 + 11 = 14$

$Sc \to e$

$Sc \to d$

$3 + 10 + 4 = 17$

$SB \to b$

$SB \to e$

$5 + 4 + 11 = 20$

$4 + 12 + 4 = 20$

$SB \to \to G$

$9 + 16 + 0 = 25$

$Sc \to d$

$3 + 7 + 6 = 16$

$Scd \to e$

$3 + 7 + 2 + 4 = 16$

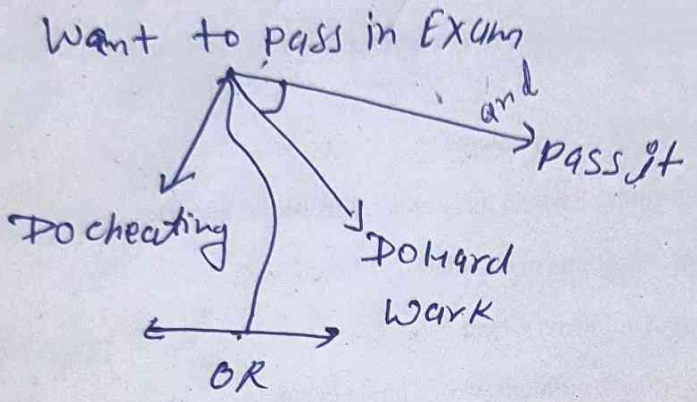
$Scde \to G$

$12 + 5 + 0 = 17$

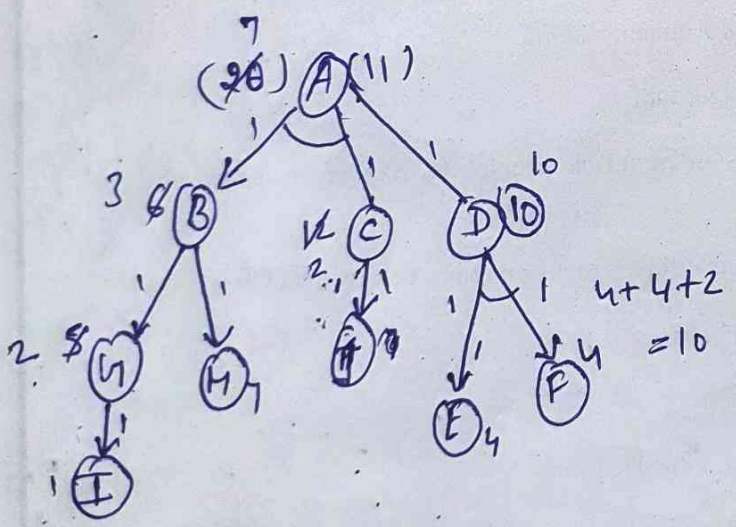
AO* Search

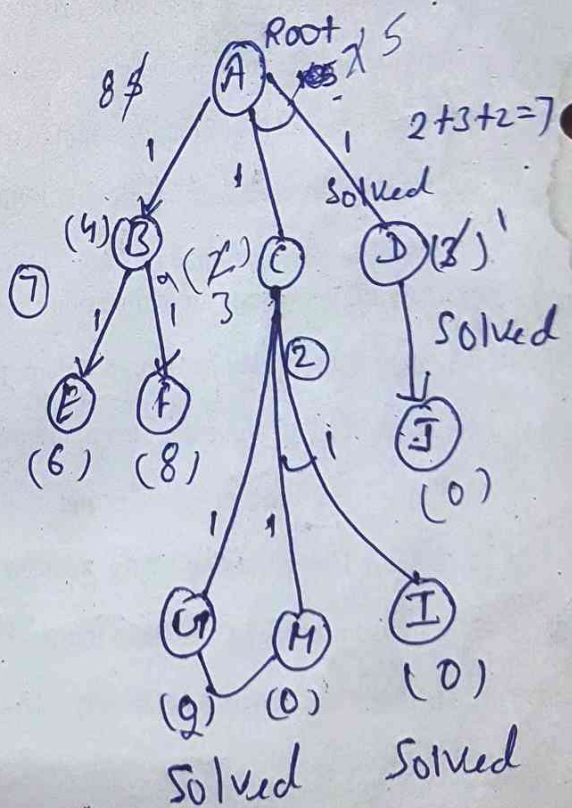
AO* (AND/OR) problem Decomposition

(Breakdown into smaller pieces)



AO* does not explore all the Solution paths once it got a Solution





Continous -
Constraint

~~Constraint~~ Constraint Satisfaction problem (CSP)

- CSP consists of three components V, D, C

→ V is set of variables $\{v_1, v_2 --- v_n\}$

- D is set of domains $\{D_1, D_2, D_3 - D_n\}$ one for each variables -

c_1 (scope, rel)

— Scope is set of variables that participate in constraint

— Rel is relation that define the ~~value~~ values that variable can take

V_1	V_2
A	B

$c_1 = (v_1 v_2), (v_1 \neq v_2)$

$c_1 = (v_1, v_2), (A, B)$

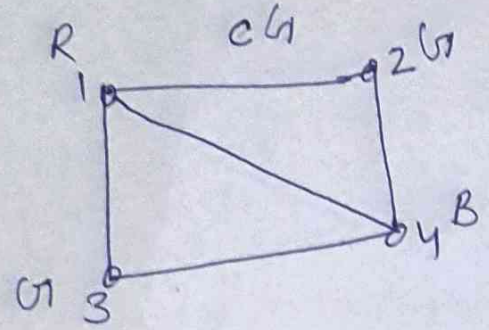
$c_1 = (v_1 v_2)$ (1,2) (1,4) (2,4)

(1,2) (2,4)

~~MAP Coloring~~

map colouring

Constraint Satisfaction problem (CSP)



$V = \{1, 2, 3, 4\}$

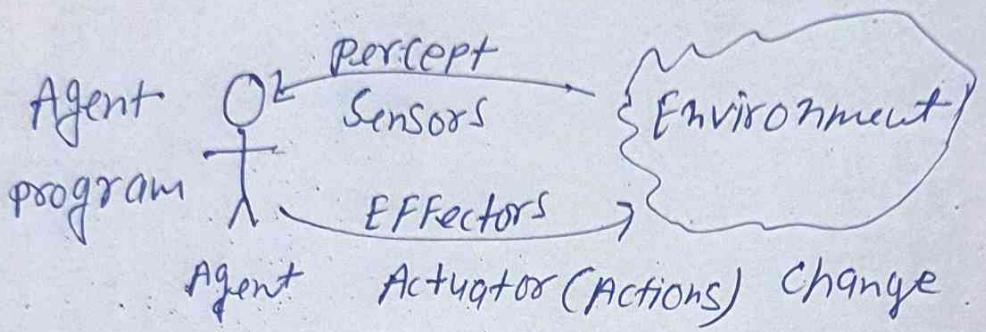
$D = \{Red, Green, Blue\}$

$C = \{1 \neq 2, 1 \neq 3, 1 \neq 4, 2 \neq 4, 3 \neq 4\}$ Backtracking

	1	2	3	4
initial dom.	R,G,B	RGB	RGB	RGB
1 = R	R	GB	GB	GB
2 = G	R	G	GB	B
3 = G	R	G	G	B

Introduction to Intelligent Agents.

Agents / Intelligent Agents



Agent → High Performance
Optimized Result
right ~~Rational~~ Action

P: Performance
E: Environment
A: Actions
S: Sensors

Types

- Simple Reflex Agents
- model Based Reflex -
- Goal Based Agents
- Utility Based Agents
- Learning Agents